Md Kaif Afran Khan

Data Engineer & AI Engineer

Dhaka, Bangladesh | +880 1971197651

kaif.khan@northsouth.edu | github.com/Afran-zero

PROFESSIONAL SUMMARY

Highly motivated Data and AI Engineer with hands-on experience building scalable data pipelines, real-time streaming architectures, and production-grade AI applications. Proficient in AWS cloud services, big data technologies (Spark, Kafka), and LLM integration. Strong background in computer science with a proven track record of deploying full-stack, data-driven solutions to solve complex business problems.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript/JavaScript, SQL, PHP, C++
Data Engineering: Apache Kafka, Apache Spark (Structured Streaming), dbt, Apache Airflow, Prefect, Medallion Architecture, ETL/ELT Pipelines
Cloud & DevOps: AWS (S3, Lambda, Glue, Athena, EventBridge, ECS/Fargate, EC2, Bedrock), Docker, Redis, CI/CD
AI & Machine Learning: LLMs (Prompt Engineering, LoRA Fine-tuning), RAG, AWS Bedrock (Claude), Scikit-learn, XGBoost, SHAP, OpenRouter
Backend & Databases: FastAPI, Flask, Node.js, MongoDB, PostgreSQL, MySQL, Supabase, Sanity CMS
Frontend: React, Next.js, Tailwind, Bootstrap, React Native (Expo)

PROFESSIONAL EXPERIENCE

AI Engineer *The Data Island* *2025 – Present*

- Architected and built an automated form-filling system for Aselo, extracting structured form data from S3 transcripts using AWS Bedrock (Claude).
- Engineered a scalable backend using FastAPI and AWS S3, with a responsive React + TypeScript frontend.
- Implemented MFA session support and real-time chat view, ensuring the architecture is serverless-ready (Lambda + API Gateway) or containerized for ECS deployment.

Core Volunteer & Tech Lead *Farzana Afroz Foundation (NGO)* *2022 – Present*

- Lead the technical development and maintenance of the organization's digital platforms and currency technology initiatives.
- Managed and scaled the foundation's official website (farhanaafrozfoundation.org), ensuring high availability and dynamic content management.

DATA ENGINEERING PROJECTS

MarketPlus – Cloud-Based Stock Analytics Pipeline
AWS S3, Athena, Glue, Spark, Prefect, FastAPI, React

- Engineered a robust Medallion Architecture (Bronze → Silver → Gold) data pipeline for processing stock market data.
- Implemented cloud ingestion using AWS EventBridge and Lambda (Alpha Vantage API) with local Spark processing orchestrated by Prefect.
- Computed advanced analytics (daily returns, moving averages, volatility, top movers) and exposed data via FastAPI for a React dashboard.

CryptoStream – Real-Time Market Data Pipeline
Kafka, Spark Structured Streaming, Airflow, dbt, Postgres, Docker

- Built an end-to-end real-time streaming pipeline ingesting WebSocket data through Kafka and Spark Structured Streaming into Postgres.
- Utilized dbt for Silver and Gold transformations, orchestrated by Airflow (running every 5 minutes).
- Generated 1-minute OHLCV candles and moving averages, complete with health monitoring and a full Docker Compose stack.

AI & SOFTWARE ENGINEERING PROJECTS

STORE – Multi-Tenant Smart ERP Platform
FastAPI, MongoDB, React, React Native, Redis, Docker

- Developed a full ERP platform with three surfaces: Backend API, Web Dashboard, and Mobile Worker App.
- Integrated an AI Assistant (OpenRouter) and enforced multi-tenancy via `businessId`. Modules include Auth/RBAC, Inventory, Sales, and Analytics.

Denz E-Commerce (Live: denzapparel.com)
Next.js 14, TypeScript, Supabase, Tailwind

- Built a modern full-stack e-commerce platform featuring Google OAuth, an admin dashboard, and secure role-based access.
- Implemented HttpOnly cookies, Row Level Security (RLS), and Cloudinary integration for media management.

RESEARCH & PUBLICATIONS

Explainable Hybrid ARIMA–ML Forecasting of Bangladesh's GDP
5th IEEE International Conference on Robotics, Automation, Artificial-intelligence and Internet-of-Things (2026)

- Developed a Hybrid Residual Learning Ensemble (ARIMA + XGBoost/RF/GBM/SVR) with SHAP explainability, achieving a best MAPE of 7.60%.

FSM-LoRA: Dynamic Gradient Gating via Finite State Machines for Energy-Efficient LoRA Fine-Tuning
2nd International Conference on AI and Emerging Technology for Sustainable Future (2026)

- Designed an FSM controller to skip redundant backward passes, reducing wall-clock time by up to 58.58% with minimal accuracy loss.

EDUCATION

North South University *Dhaka, Bangladesh*
B.Sc. in Computer Science & Engineering *Expected 2026*
CGPA: 3.62/4.00 | Distinction: Cum Laude | 25% Merit Scholarship

Dhaka Residential Model College *Dhaka, Bangladesh*
Higher Secondary Certificate (HSC) & Secondary School Certificate (SSC)

CERTIFICATIONS & LANGUAGES

Certifications: AWS Data Engineer Practitioner (Coursera), Full Stack Developer (MERN Stack), Data Science Short Course (Stanford Online).

Languages: English (IELTS 7.5), Bangla (Native).